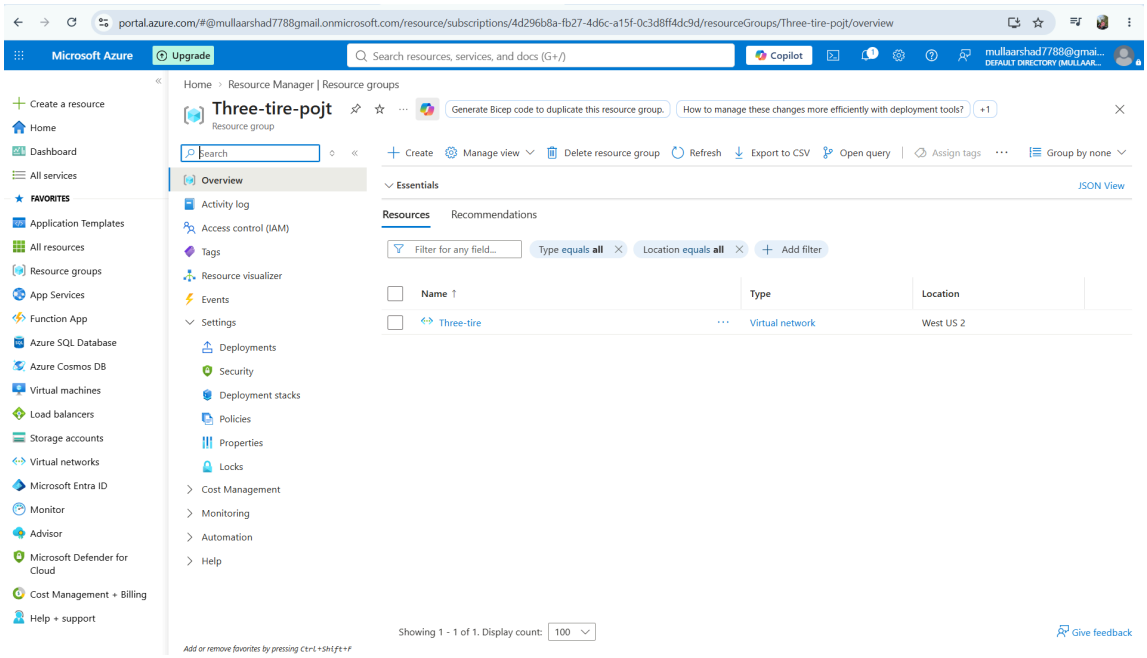
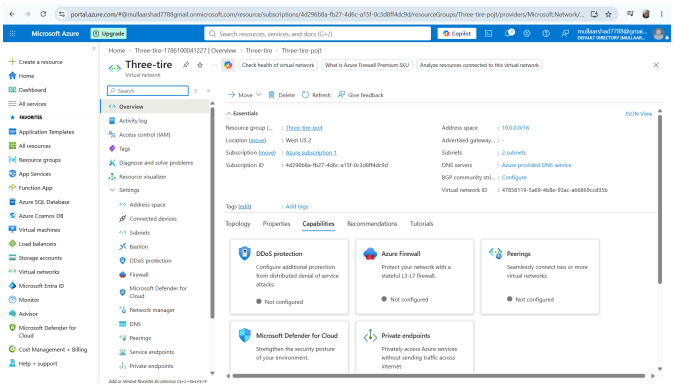
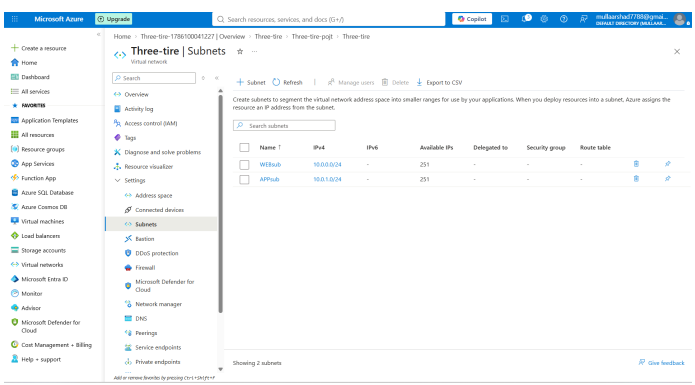
Three-Tier Architecture

Firstly, I logged into the Azure portal, and created a resource group named Three-tire-pojt.



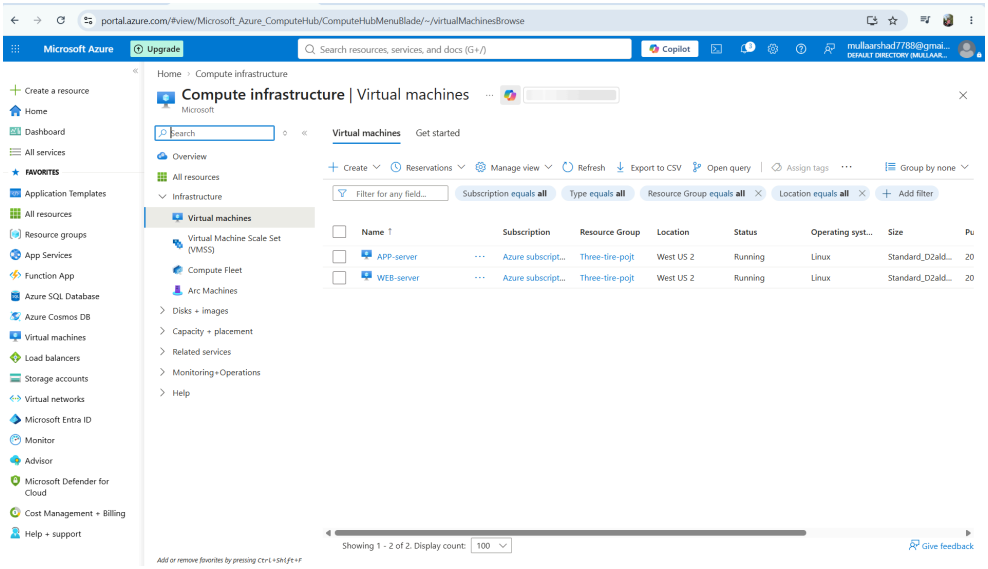
Also, created a Vnet and Two subnets in it Named as Three-tire(VNet), WEBsub(subnet1), APPsub(subnet2).





Created two VM’s (Virtual Machines),

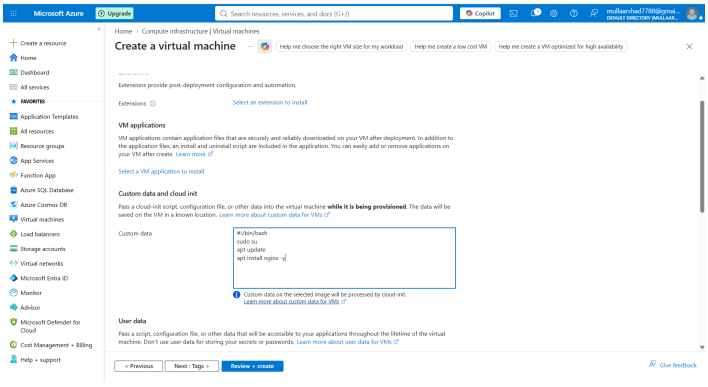
1. WEB-server (VM1)
2. APP-server (VM2)



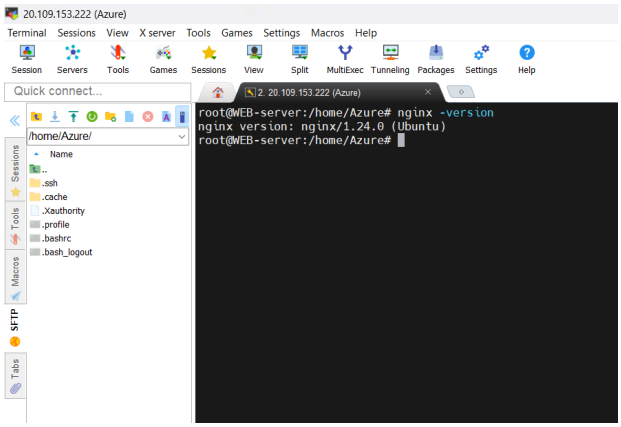
While creating VM’s i have chosen the Ubuntu images to run my machines and install nginx on both machines In Advanced settings.

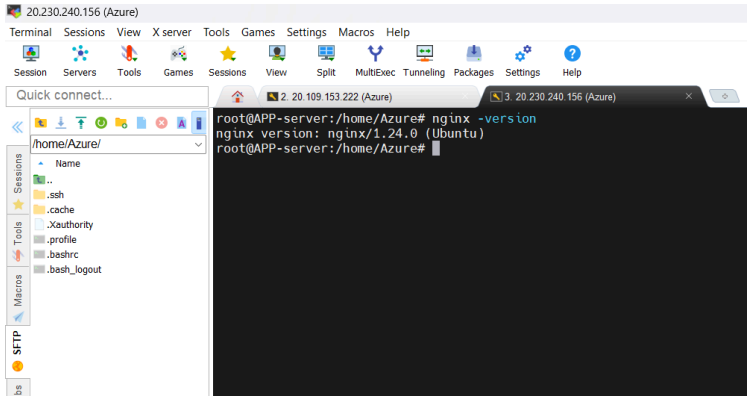
With a simple command,

```
#!/bin/bash
sudo su
apt update
apt install nginx -y
```



And, I have logged in my servers with a public IP address on Mobaxtrem, my servers are successfully UP and Running installed nginx on both servers.





On APP-server i have installed Tomcat with this commands;

Step 1: Update the system

sudo apt update
sudo apt upgrade -y

Step 2: Install Java

Tomcat requires Java.

sudo apt install openjdk-21-jdk -y

Step 3: Create a Tomcat user

sudo groupadd tomcat
sudo useradd -s /bin/false -g tomcat -d /opt/tomcat tomcat

Step 4: Download Tomcat

Move to the temporary directory:

```
cd /tmp
wget https://archive.apache.org/dist/tomcat/tomcat-10/v10.1.46/bin/apache-tomcat-10.1.46.tar.gz
```

Step 5: Extract Tomcat

Create the installation directory:

sudo mkdir -p /opt/tomcat

Extract:

sudo tar -xzf apache-tomcat-10.1.46.tar.gz -C /opt/tomcat --strip-components=1

Step 6: Set permissions

sudo chown -R tomcat:tomcat /opt/tomcat
sudo chmod -R u+x /opt/tomcat/bin

Step 7: Find your Java path

readlink -f $(which java)

Step 8: Create the Tomcat service

sudo nano /etc/systemd/system/tomcat.service
[Unit]
Description=Apache Tomcat Web Application Container
After=network.target

[Service]
Type=forking

User=tomcat
Group=tomcat

Environment=JAVA_HOME=/usr/lib/jvm/java-21-openjdk-amd64
Environment=CATALINA_HOME=/opt/tomcat
Environment=CATALINA_BASE=/opt/tomcat
Environment=CATALINA_PID=/opt/tomcat/temp/tomcat.pid

ExecStart=/opt/tomcat/bin/startup.sh
ExecStop=/opt/tomcat/bin/shutdown.sh

Restart=always

[Install]
WantedBy=multi-user.target

Step 9: Reload systemd

sudo systemctl daemon-reload

Enable Tomcat:

sudo systemctl enable tomcat

Start Tomcat:

sudo systemctl start tomcat

Step 10: Check the status

sudo systemctl status tomcat

Step 11: Open port 8080

sudo ufw allow 8080/tcp

Reload:

sudo ufw reload

Step 12: Test Tomcat

Check locally:

curl http://localhost:8080

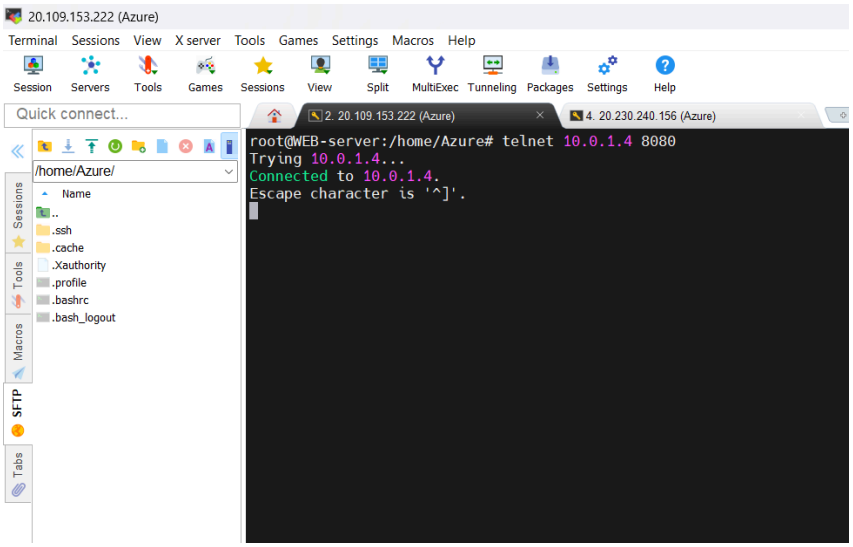
In WEB-server i have write a rule in inbound rules where,

source= any
destination= WEB-server public IP
service= http
Destination port= 80
Action= Allow

And in APP-server the rule i wrote to connect form web server to app server is,

source= WEB-server public IP
destination= APP-server public IP
service= costum
Destination port= 8080
Action= Allow

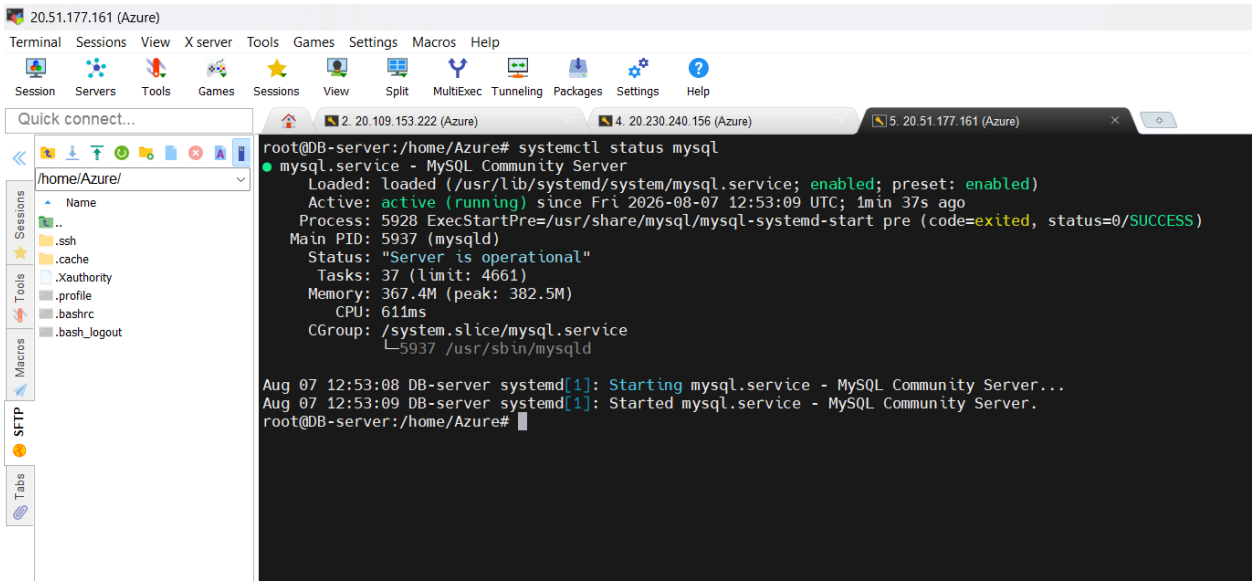
Command to connect WEB-server to APP-server
telnet <APP IP> 8080
Example: telnet 10.0.1.4 8080
I have successfully connected to my APP-server form WEB-server



Later, I created another VM (DB-server).
The reason for creating another VM later is, I am using a free trial account in azure. In azure you can't create 3 VM's in a single

region, so I have chosen the west US region for DB-server. While the both WEB & APP servers are created in east US2.
Now in DB-server I have added an ubuntu image and installed nginx while creating a VM.

In DB-server i installed MySql



MySql is successfully installed and running in DB-server

Commands i used to install DB-server:

Step 1:
Sudo su
apt update

Step 2:
apt install mysql-server –y

Step 3:
Mysql_secure_installation

Step 4:
nano /etc/mysql/mysql.conf.d/mysqld.cnf
And edited this bind-address = 0.0.0.0

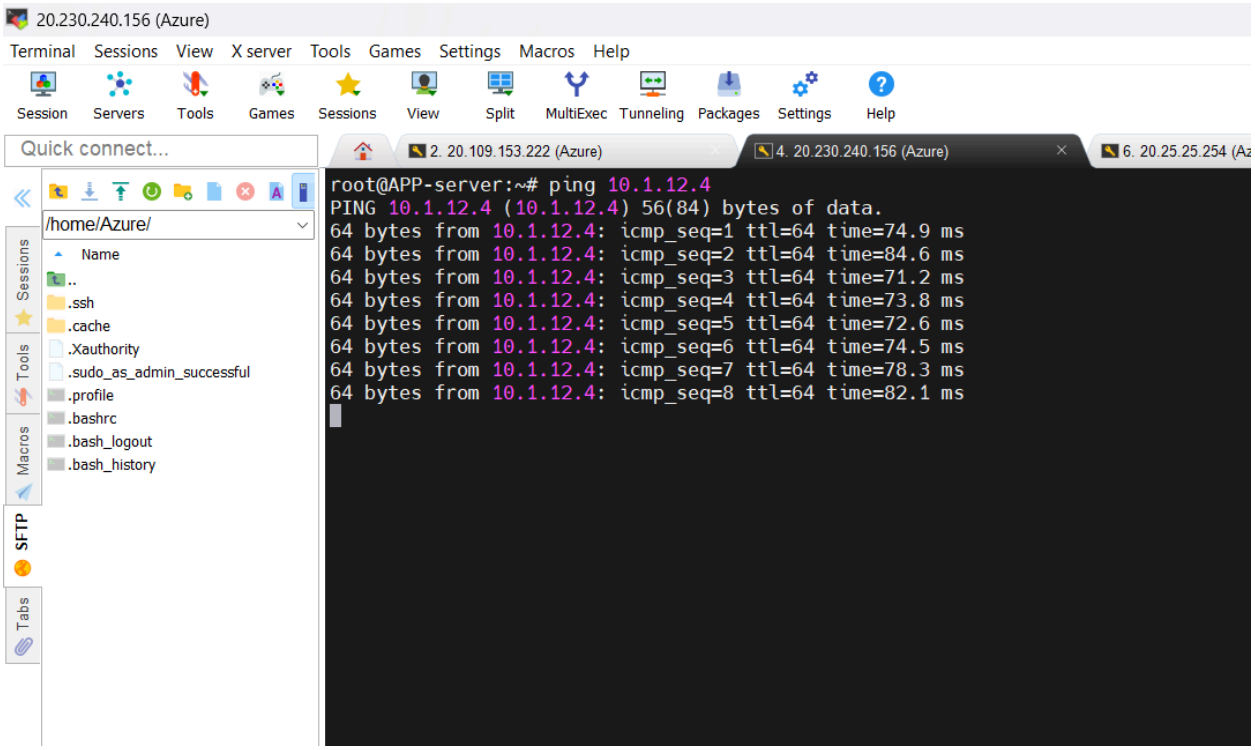
Step 5:
systemctl restart mysql

Step 6:
systemctl status mysql

Now i created a VNet peering to connect the APP-server and DB-server.

I have made sure that my VNet IP addresses are not overlapped.

Successfully Established peering connection between WEB to APP servers.
To the check the connection i have used ping command
Example: ping 10.1.12.4



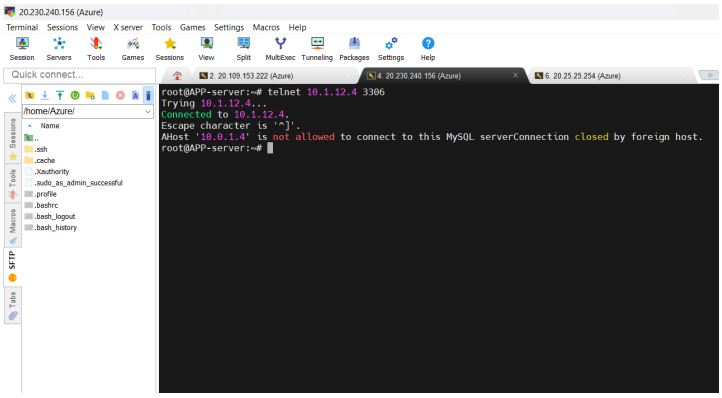
In, DB-server we need to write the inbound rules to make connected status for APP to DB.

The rule are;
source= APP-server public IP
destination= DB-server public IP
service= costum
Destination port= 3306
Action= Allow

Also i have write another rule for WEB to DB;
source= WEB-server public IP
destination= DB-server public IP
service= costum
Destination port= Any
Action= Deny

This makes the traffic form WEB to DB are blocked and allows the traffic for APP to DB servers.

Then I checked the connection status for the APP to DB server, and i have successfully connected to DB-server.



The three-tire-arcitrature is successfully completed by connecting WEB to APP & APP to DB servers.

MULLA.ARSHAD AALEM

PROJECT COMPLITION DOCUMENT